PROJECT REPORT

GLOBAL FUEL PRICES AND TAXATION TRENDS (2020–2026)

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Project domain: Data Visualisation & Analytics | Platform: Tableau

Project Title	Global Fuel Prices and Taxation Trends (2020–2026)
Student	Anushka Dubey
Dataset Scope	84 countries | 7 regions | 7,056 quarterly records
Fuel Categories	Petrol, Diesel and LPG
Technology	Tableau (dashboard and visual analytics)
Submission Date	02 November 2026

Academic project report

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1. INTRODUCTION

1.1 Project Overview

Global Fuel Prices and Taxation Trends is a data visualisation project designed to explore how petrol, diesel and LPG prices vary across countries and regions over time. The project uses a quarterly dataset covering 84 countries across 7 regions from 2020 to 2026, containing 7,056 records. Tableau is used to organise the information into an interactive dashboard with charts, summary indicators and filters.

The dashboard brings several related perspectives into one place: price movements over time, regional and country comparisons, estimated taxation patterns, and the relationship between subsidies and fuel prices. It is intended for academic exploration and communication of trends.

1.2 Purpose

The purpose is to make a large fuel-price dataset easier to explore and interpret. Instead of manually comparing separate tables or charts, a user can select a year, region or fuel type and review the corresponding visual summaries. The project also demonstrates the use of data preparation, analytical thinking and dashboard design.

2. IDEATION PHASE

2.1 Problem Statement

Fuel prices differ across countries and change over time. These differences may be associated with factors such as taxation, subsidies, market conditions and the type of fuel. When information is spread across multiple tables and sources, users may find it difficult to compare countries consistently, identify trends or communicate findings clearly.

2.2 Empathy Map Canvas

Target persona: A student, researcher or early-career analyst who needs to understand global fuel-price patterns for an academic task or introductory market analysis.

SAYS	THINKS
“I need to compare fuel prices across countries.” “I want to see how prices changed over time.”	Which regions have higher prices? Are tax and subsidy patterns related to price differences? Can I trust the figures and compare them fairly?
DOES	FEELS
Searches for data, reviews charts, compares countries and uses filters to explore selected periods.	May feel confused by scattered data; feels more informed when trends are presented clearly and consistently.

2.3 Brainstorming

Ideas considered for the dashboard include KPI cards for high-level context; annual and quarterly trend charts; regional and country comparisons; tax-related views; subsidy comparisons; and filters for year, region and fuel type. The selected direction combines these features into a single Tableau dashboard.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	User Activity	Need / Pain Point	Dashboard Response
Discover	Looks for global fuel-price information	Information is spread across sources	One consolidated dashboard
Explore	Selects fuel type, year or region	Large dataset can be difficult to navigate	Interactive filters
Compare	Compares countries and regions	Different chart formats can make comparisons difficult	Consistent comparison views
Interpret	Reviews tax and subsidy patterns	Relationships may not be obvious	Tax and subsidy visualisations
Communicate	Shares findings in a report or presentation	Static raw tables are hard to explain	KPI cards and clear charts

3.2 Solution Requirement

Requirement Type	Requirement
Functional	Display fuel-price trends over time; compare regions and countries; show tax and subsidy analysis; provide year, region and fuel-type filters; display summary KPIs.
Data	Use consistent country, region, date, fuel-type and numeric measure fields. Check data types, missing values, duplicates and units before visualisation.
Usability	Use readable titles, legends, labels and a consistent layout. Filters should be easy to locate and apply.
Reliability	Charts and KPI values should respond consistently to filters. Calculations should be checked against the underlying data.

3.3 Data Flow Diagram

1. Dataset Quarterly fuel-price records	**2. Data Preparation** Check fields, data types, missing values and consistency	**3. Tableau Data Model** Connect data and define measures / dimensions	**4. Visualisations** Build charts, KPIs and filters	**5. Dashboard Output** Explore and communicate insights

Data moves from the source dataset through preparation and Tableau, then into the visualisations and final dashboard. User filter selections change the displayed view without changing the source dataset.

3.4 Technology Stack

Component	Technology / Resource	Purpose
Visualisation	Tableau Public / Tableau Desktop	Build interactive charts and dashboard
Data source	Project quarterly fuel-price dataset	Country, region, date, fuel type, price, tax and subsidy fields
Preparation	Spreadsheet / data preparation tools	Inspect, clean and organise data before analysis
Documentation	PDF and presentation tools	Record methodology, testing and project findings

4. PROJECT DESIGN

4.1 Problem-Solution Fit

Problem: Users need a simple way to compare global fuel prices and explore differences across time, countries, regions and related tax/subsidy measures.

Target users: Students, academic researchers, business learners and users interested in energy-market data.

Value proposition: A single interactive Tableau dashboard that turns quarterly records into accessible visual comparisons and trend summaries.

4.2 Proposed Solution

The proposed solution is an interactive Tableau dashboard with nine analytical views and supporting KPI cards. The dashboard is designed to let users explore the data through year, region and fuel-type filters.

No.	Dashboard View	Purpose
1	Fuel Price Trends	Summarises fuel-price movement across the selected period.
2	Quarterly Fuel Price Trends	Shows quarter-by-quarter changes and fluctuations.
3	Regional Comparison	Compares fuel-price patterns across regions.
4	Regional Tax Comparison	Compares the tax-related measure across regions.
5	Country Ranking	Ranks countries by the selected fuel-price measure.
6	Subsidy vs Fuel Price	Explores the relationship between subsidy values and fuel prices.
7	Tax Analysis	Provides a closer view of tax measures in the dataset.
8	Tax Share by Income Group	Compares tax share across income groups, where the required field is available.
9	Taxation vs Subsidies	Places taxation and subsidy measures in comparative context.

4.3 Solution Architecture

Data Layer	Quarterly country-level fuel-price dataset, including fuel category and related tax/subsidy fields.
Preparation Layer	Field validation, type checking, data cleaning and calculated measures where required.
Visualisation Layer	Tableau worksheets, KPI cards, dashboard layout, legends and filters.
Presentation Layer	Interactive dashboard, exported screenshots/PDF and project demonstration.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The work is organised into four illustrative sprints. The schedule below is a planning baseline and should be updated to reflect the actual dates and completion status of the project.

Sprint	Focus	Key Tasks	Planned Duration
1	Data preparation	Understand fields; check data quality; prepare the data source; connect to Tableau.	6 days
2	Analysis	Create trend, regional, country, tax and subsidy views.	6 days
3	Dashboard build	Arrange nine views; add KPI cards; configure filters; refine labels and layout.	6 days

Sprint	Focus	Key Tasks	Planned Duration
4	Testing & delivery	Test interactions and calculations; document results; prepare demo and final report.	6 days

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Testing should confirm that the dashboard is accurate, usable and responsive. The following cases provide a practical test checklist. Record actual results on the device used for submission; no measured load-time or benchmark values are assumed in this report.

Test ID	Test Case	Expected Result	Status
T01	Open the Tableau workbook and dashboard	Workbook opens and dashboard renders without errors.	To be recorded
T02	Select a year filter	All relevant charts and KPIs update to the selected year.	To be recorded
T03	Select a region filter	Regional selection is reflected consistently across applicable views.	To be recorded
T04	Select a fuel type	Views show the selected fuel category and labels remain clear.	To be recorded
T05	Check KPI values against source data	Displayed values match the intended aggregation and filter context.	To be recorded
T06	Inspect chart labels and legends	Titles, units, labels and legends are readable and unambiguous.	To be recorded
T07	Test dashboard on presentation device	Dashboard is usable at the intended screen size and does not show clipped content.	To be recorded

Performance considerations

Evaluate initial workbook opening, filter response, chart rendering and usability on the actual presentation device. If performance is slow, reduce unnecessary marks, simplify calculations, limit unused fields and optimise dashboard layout.

7. RESULTS

7.1 Output Screenshots

Insert screenshots of the completed Tableau dashboard in this section before final submission. Recommended evidence includes: (1) full dashboard view, (2) fuel-price trend view, (3) regional/country comparison, (4) tax and subsidy analysis, and (5) an example showing filters applied.

7.2 Result Summary

The project produces an interactive dashboard containing nine visualisation views, KPI cards and filters for year, region and fuel type. It supports exploration of fuel-price trends, geographic differences and dataset-based tax/subsidy patterns. Specific findings and numerical conclusions should be added after checking the final dashboard and selected filter context.

8. ADVANTAGES & DISADVANTAGES

Advantages	Disadvantages / Limitations
• Consolidates multiple analytical views in one dashboard. • Interactive filters support focused exploration. • Visual trends and comparisons are easier to communicate than raw tables. • Demonstrates practical Tableau and data-visualisation skills.	• Findings depend on the quality and completeness of the input dataset. • Estimated/simulated values should not be treated as verified live prices. • Country comparisons may be affected by differences in units, currencies or definitions. • The dashboard does not by itself establish causal relationships between taxes, subsidies and prices.

9. CONCLUSION

The Global Fuel Prices and Taxation Trends project demonstrates how a multi-country quarterly dataset can be transformed into an interactive Tableau dashboard. By combining trend analysis, regional and country comparisons, and tax/subsidy views, the project makes it easier to explore the dataset and communicate patterns. The dashboard is an academic analytical tool; its conclusions should be interpreted within the scope and limitations of the supplied data.

10. FUTURE SCOPE

- Replace or supplement academic estimated values with verified, regularly updated data from reputable sources.
- Add currency conversion and standardised units to improve cross-country comparisons.
- Include additional years, countries and fuel categories as reliable data becomes available.
- Add more detailed explanatory notes and source citations for each measure.
- Publish an accessible dashboard and collect feedback from users.

11. APPENDIX

Source Code	Add repository URL if source files are available; otherwise write “Not applicable – Tableau workbook project.”
Dataset Link	Insert the exact dataset source URL used in the project.
GitHub Link	Insert the project repository URL, if available.
Project Demo Link	Insert the Google Drive / video / Tableau Public link, if available.
Supporting Evidence	Attach dashboard screenshots, data dictionary and any relevant project documentation.

Important data note: *The project dataset has been described as containing estimated/simulated values for academic analysis. Clearly label it as such and do not present the dashboard as a source of verified real-time fuel prices.*